

Insights on the Current State of Reinforcement Learning in Finance

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Agenda

- Key Insights
- Preliminaries
- Evidence for Insight #1
- Evidence for Insight #2
- Conclusion
- What can we do differently?



Key Insights

- 1) RL in Finance is focused on **model-free algorithms**, applied to **equity markets** for **allocation tasks** and evaluated on **simulated backtests**.
- 2) RL in Finance suffers from recurring methodological weaknesses in **code and data sharing**, reporting of **hyperparameter tuning** and reliance on train-test paradigm for generalizability despite potential **non-stationarity** of financial markets.



Preliminaries - Data

- 543 query results in Scopus
- 163 studies after screening
 - 160 experimental
 - 3 literature review

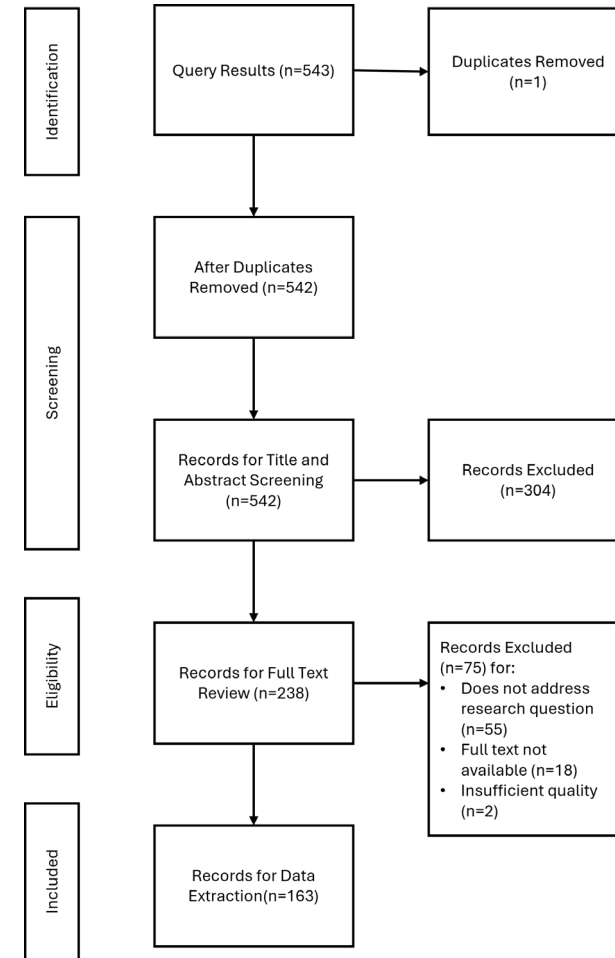
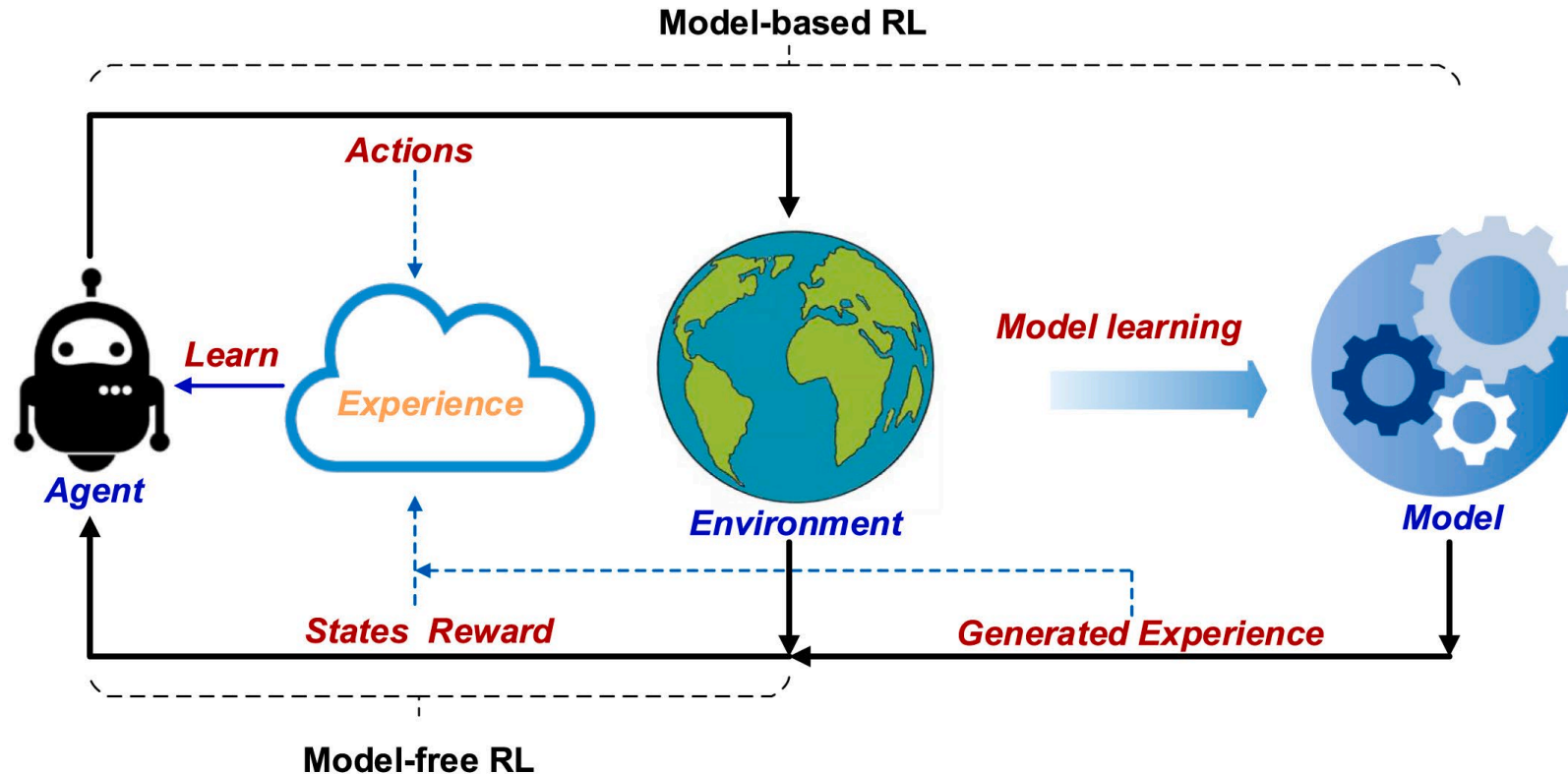


Fig 1: PRISMA-style flow diagram visualizing the selection process. We included 163 records in the final corpus after the screening process.

Preliminaries – Reinforcement Learning



[Comparative study of model-based and model-free reinforcement learning control performance in HVAC systems - ScienceDirect](#)

Preliminaries - Applications

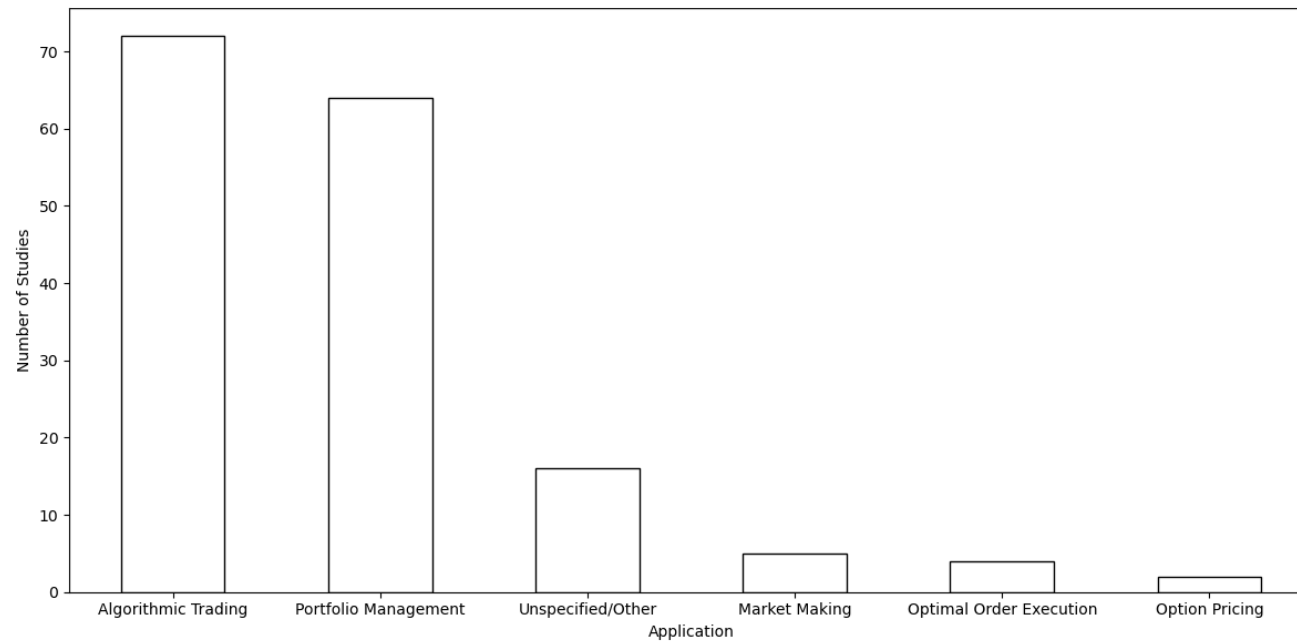


Fig 3: Number of studies by application [1]. We can see that Algorithmic Trading and Portfolio Management are the most common applications. The categories are not mutually exclusive, therefore a study might be counted in multiple categories.

Insight # 1

RL in Finance is focused on

- **model-free algorithms,**
- applied to **equity markets**
- for **allocation tasks** and
- evaluated on **simulated backtests.**



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Insight #1 – Model Free Algorithms

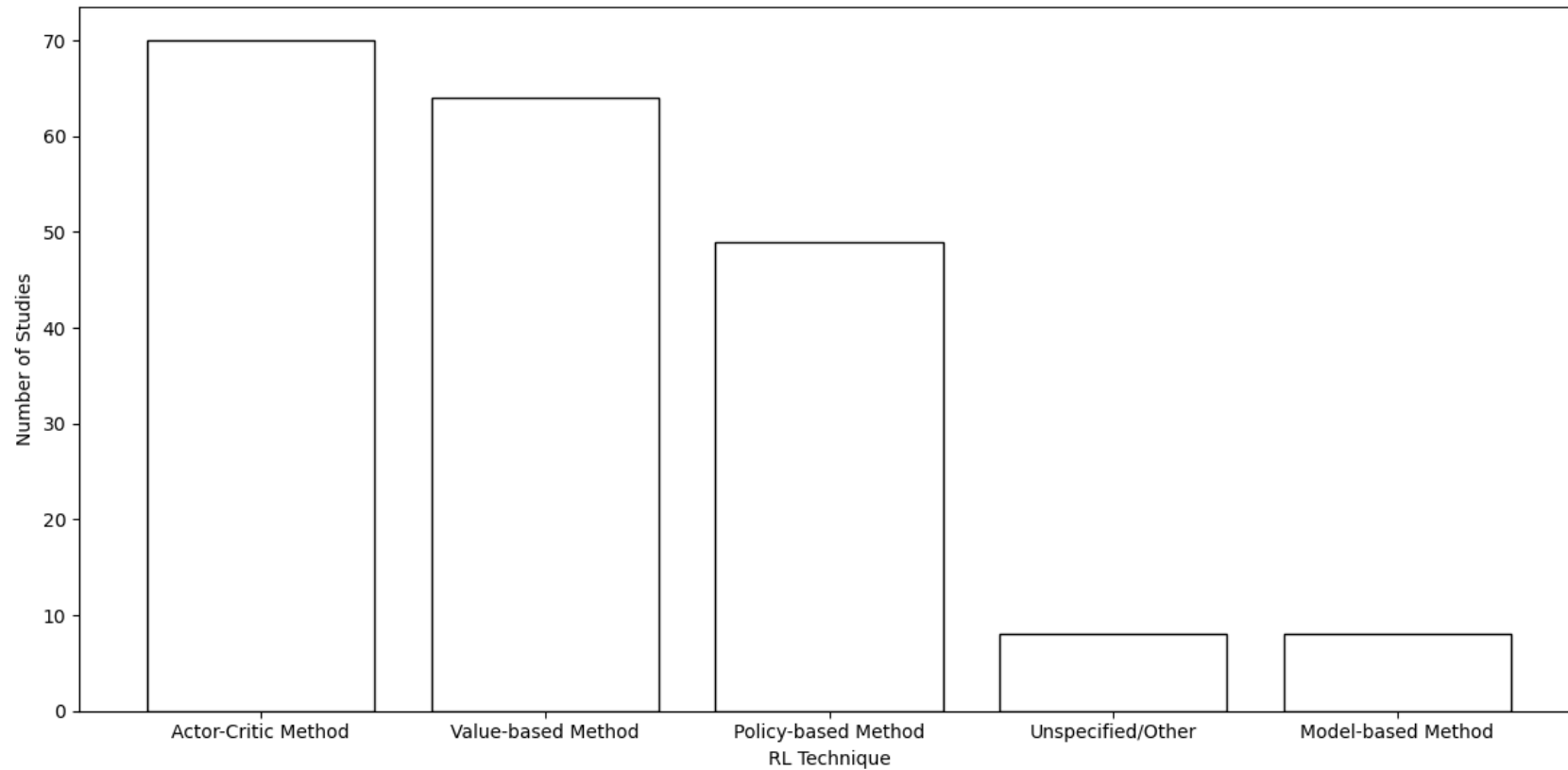


Fig 4: Number of studies by reinforcement learning technique. We can see that model-free RL methods are most common.

Insight #1 – Equity Markets

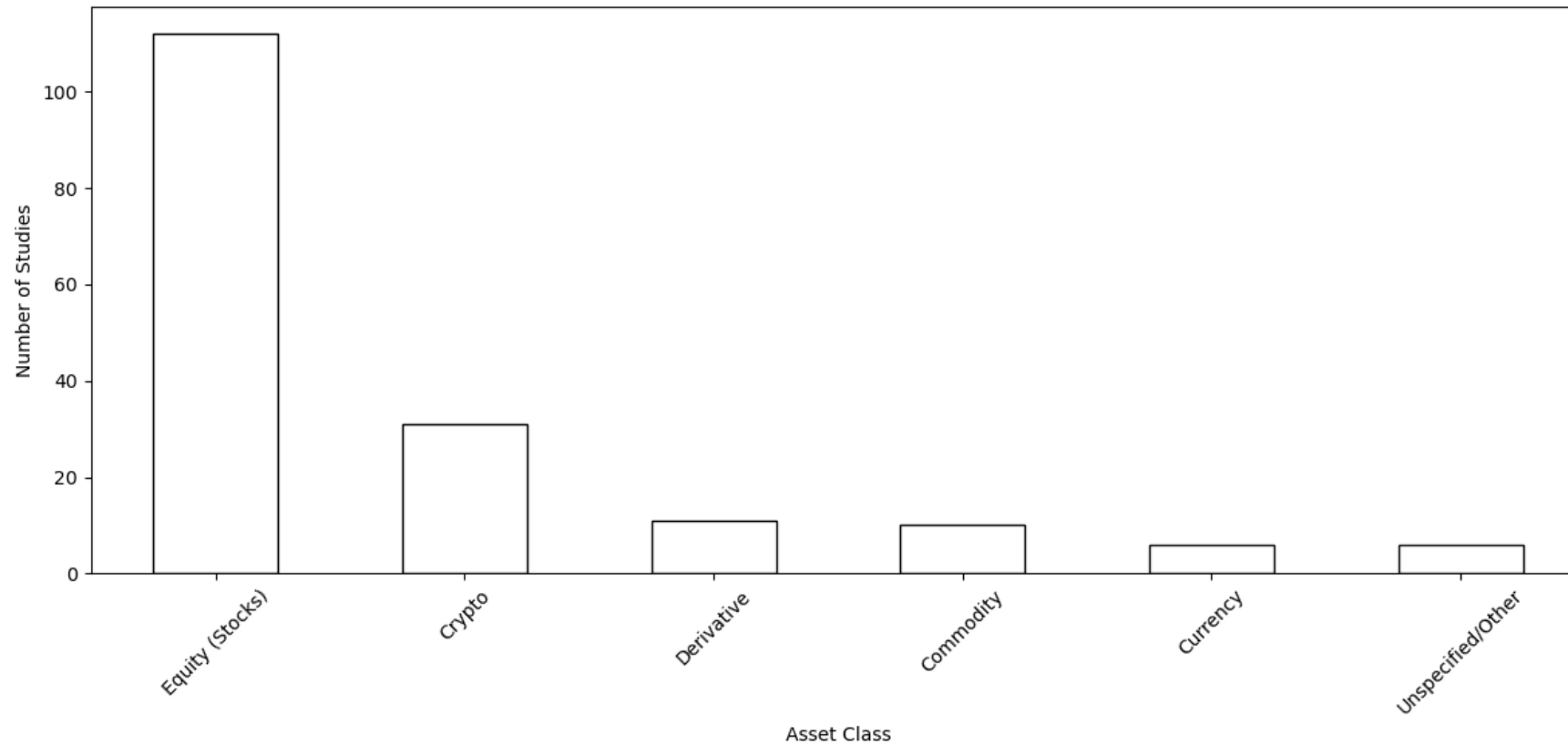


Fig 5: Number of studies asset class. We can see equity is most common.

Insight #1 – Allocation Task

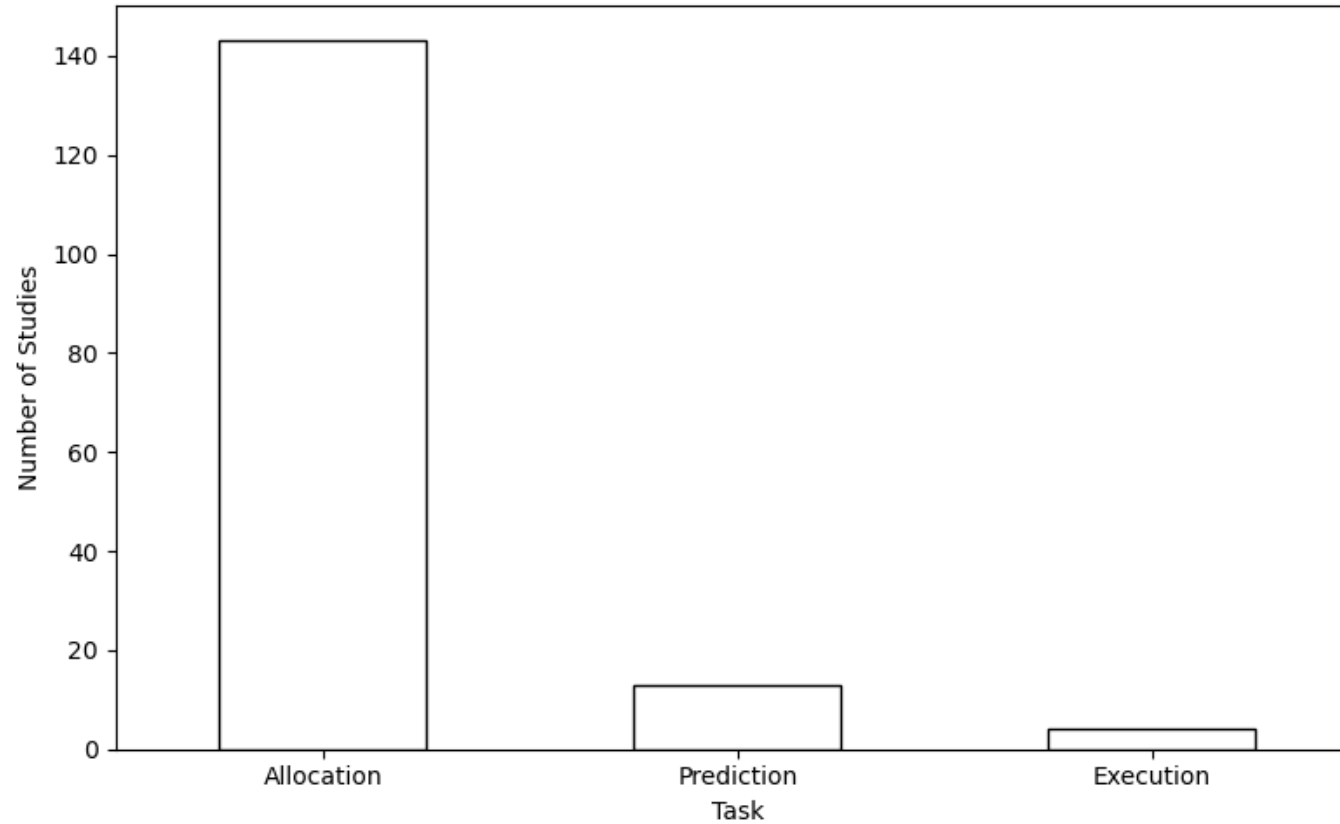


Fig 6: Number of studies by task. We can see that the allocation task is most studied.

Insight #1 – Implementation Status

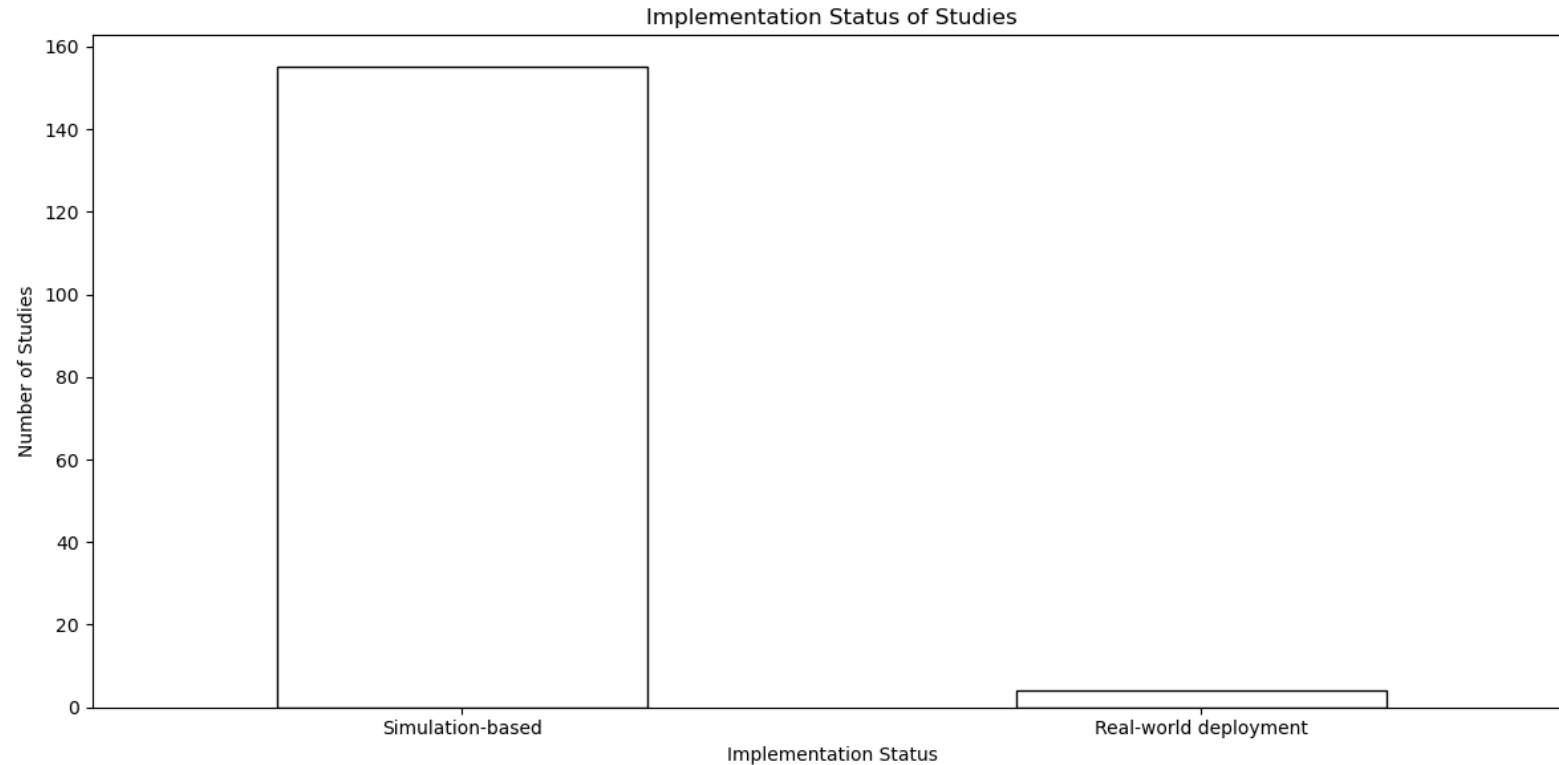


Fig 7: Number of studies by implementation status. We can see that simulation is most common with almost no real-world deployment.

Insight #2

RL in Finance suffers from recurring methodological weaknesses in

- **code and data sharing**,
- reporting of **hyperparameter tuning** and
- reliance on train-test paradigm for generalizability despite potential **non-stationarity** of financial markets.



Insight #2 – Code and Data Sharing

Reproducibility Criterion	Count	Of 17 Code-Sharing Studies	Of 160 Total Studies
Re-run as is (same code + data)	14	82.4 %	8.8 %
Re-run with updated price data (same code + updated data)	2	11.8 %	1.3 %
Re-run with custom data (same code + custom data)	12	70.6 %	7.5 %
Code available but broken/missing/misleading	2	11.8 %	1.3 %

Fig 8: Summary of reproducibility across 17 code-sharing studies.

Insight #2 – Hyperparameter Tuning

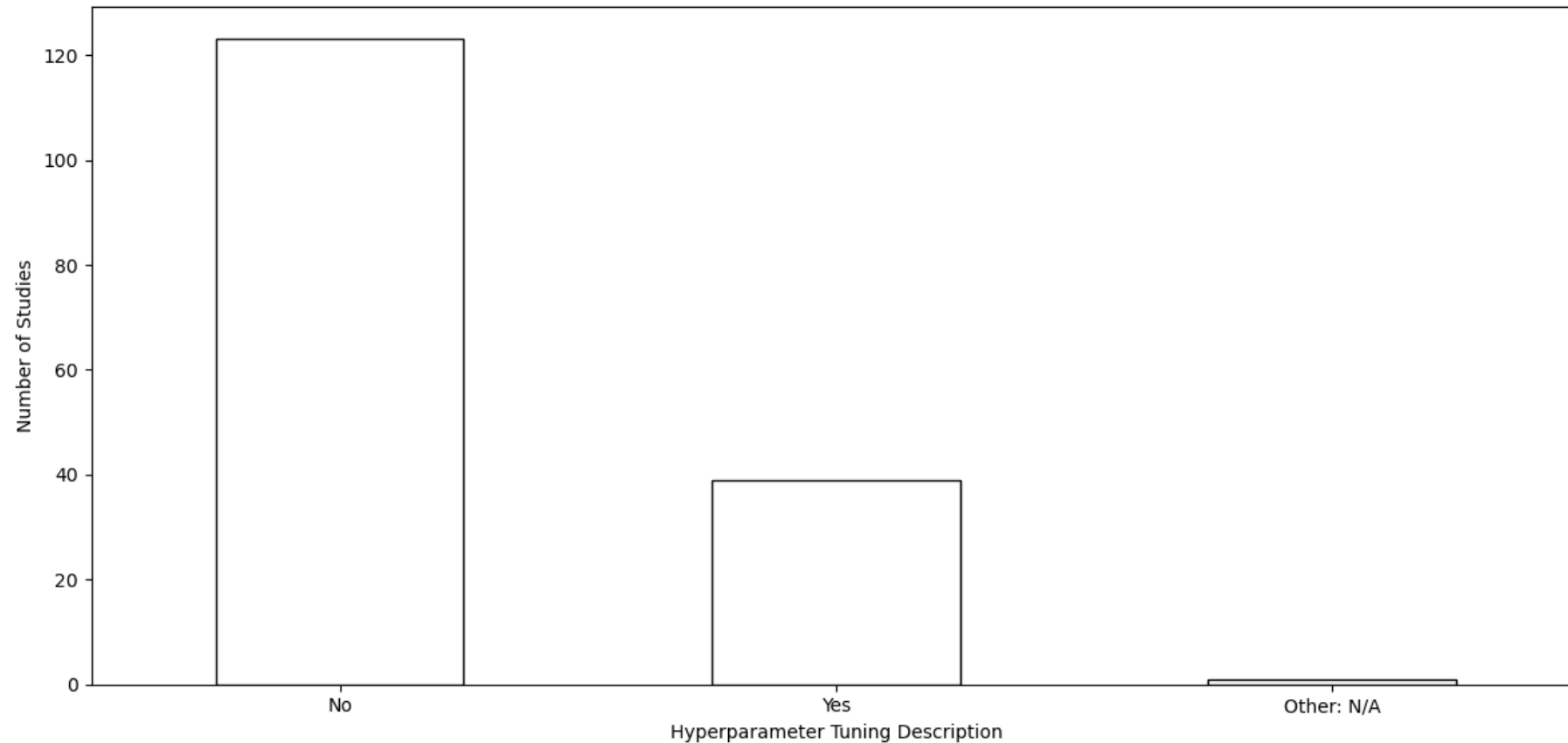


Fig 9: Number of studies explaining hyperparameter tuning process. We can see that most studies do not explain the applied hyperparameter tuning process.

Insight #2 – Non-Stationarity

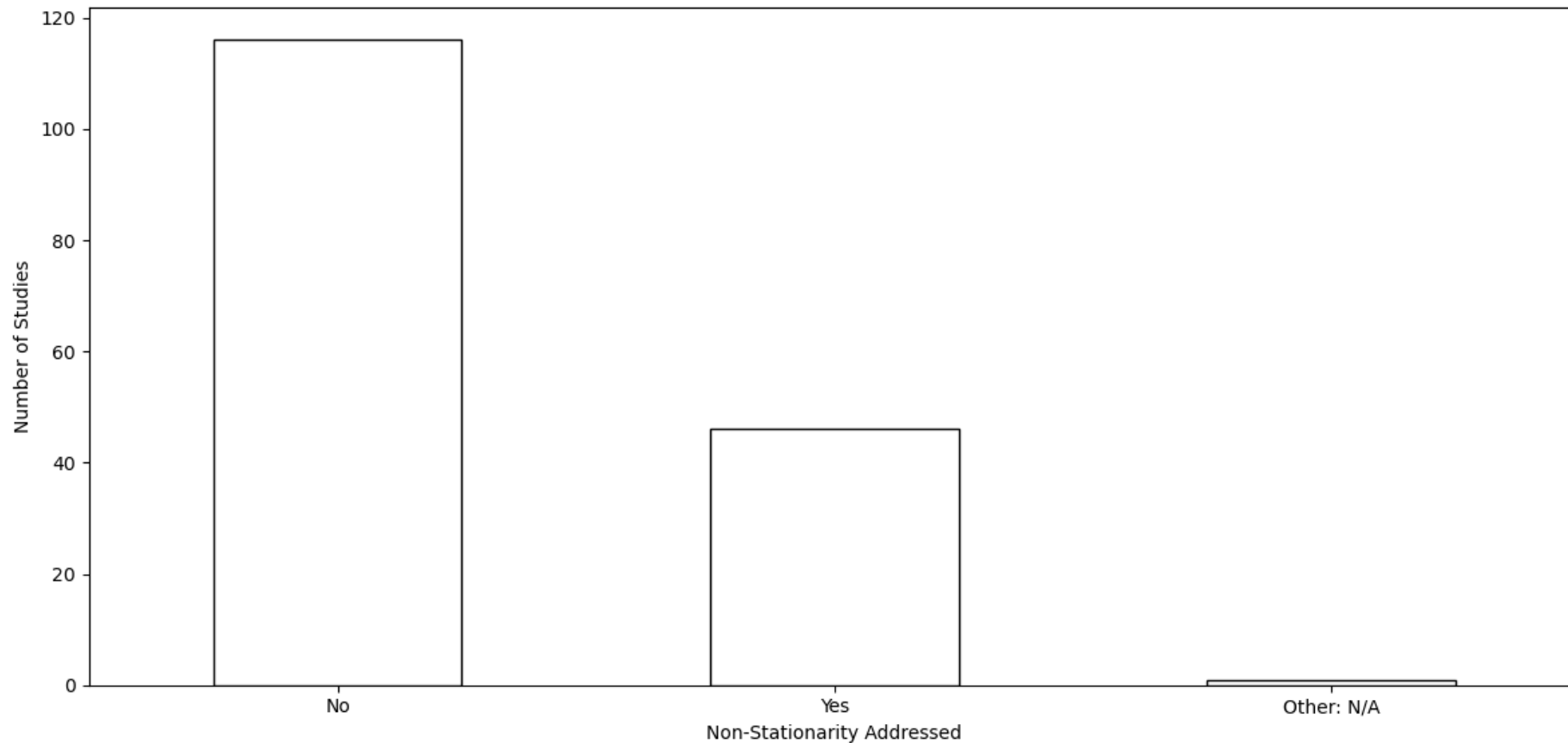


Fig 10: Number of studies addressing non-stationarity. We can see that most studies do not address non-stationarity and its implications for generalizability.

Conclusion

- Experimentally active but methodologically stagnant research field [2].
- Literature is fragmented and not cumulative.
- Methodological progress seems necessary to advance.



What can we do differently?

- 1) Improve Reproducibility
- 2) Standardize Tasks
- 3) Standardize Reporting
- 4) Standardize Datasets
- 5) Increase Realism
- 6) Introduce Taxonomy for Research Field



Example: Deployment Ladder

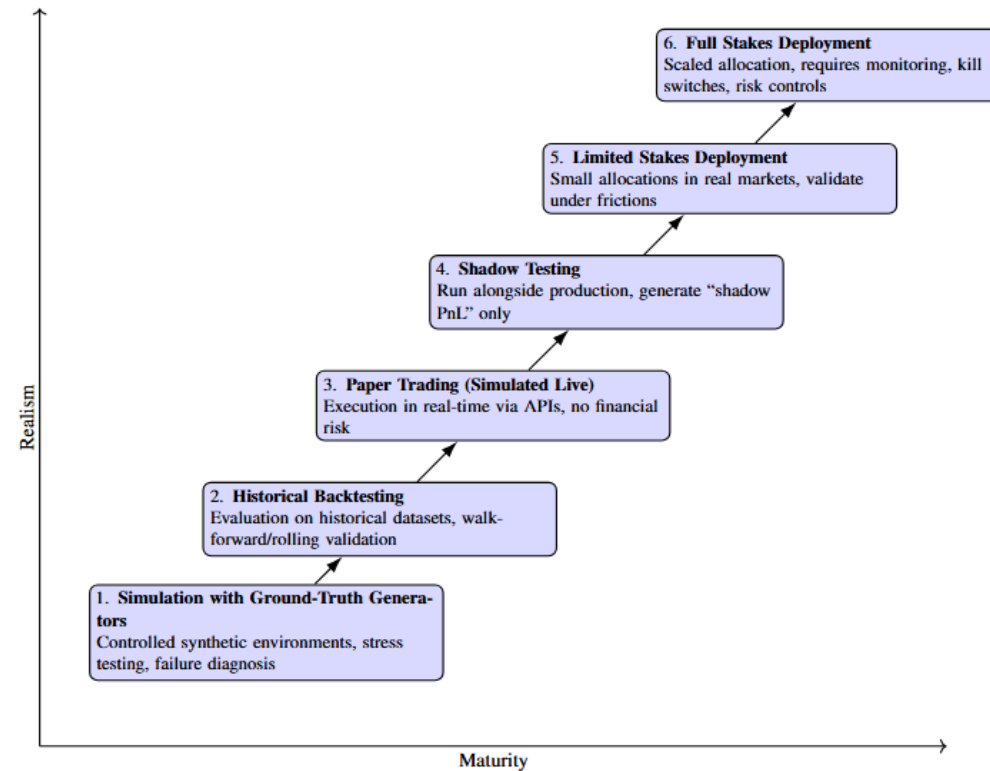


Fig 11: Deployment Ladder for RL in Finance [3]. The framework increases both maturity (x-axis) and realism (y-axis) as models progress from simulation to live trading.



References

- 1) Hambly, B., R. Xu, and H. Yang. "Recent Advances in Reinforcement Learning in Finance." *Mathematical Finance* 33, no. 3 (2023): 437–503. <https://doi.org/10.1111/mafi.12382>.
- 2) Moody, J., and M. Saffell. "Learning to Trade via Direct Reinforcement." *IEEE Transactions on Neural Networks* 12, no. 4 (2001): 875–89. <https://doi.org/10.1109/72.935097>.
- 3) Jander, Mathis and van Heeswijk, Wouter and Mes, Martijn R.K., Reinforcement Learning in Finance: From Current Practice to Standards and Frameworks. Available at SSRN: <https://ssrn.com/abstract=5703882> or <http://dx.doi.org/10.2139/ssrn.5703882>

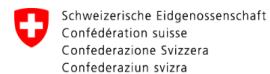


Thank you for your attention! Questions?

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This project has received funding from the Horizon Europe research and innovation programme under the Marie Skłodowska-Curie Grant Agreement No. 101119635



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